



RESEARCH

Current status of coffee (*Coffea arabica* L.) technology dissemination in the Afalach watershed, Ethiopia

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Abstract

Coffee is one of the most important global agricultural commodities and plays a key economic role in Ethiopia. However, productivity remains low in many coffee-growing areas due to limited use of improved technologies. This study assessed the current status of coffee production technologies in the Afalach watershed. From 113 total population 73 farmers were randomly selected based on their willingness and involvement in coffee production. Data were collected through semi-structured questionnaires, field observations, transect walks, key informant interviews, and focus group discussions. Both quantitative and qualitative data were analyzed using SPSS software. The results showed that most farmers had only 1–10 years of experience in coffee production, indicating that coffee farming is relatively recent in the area. The majority of households allocated less than 0.125 ha of land to coffee and produced yields below the national average (1 quintal per hectare). Most farmers relied on local coffee varieties and did not apply organic or inorganic fertilizers. Selective hand-picking was the dominant harvesting method. The study highlights limited adoption of improved coffee technologies in the watershed. Promoting improved coffee varieties, strengthening farmer training on production practices, and encouraging compost preparation are recommended to enhance productivity.

Keywords Afalach · Coffee · Hand-picking · Technology · Watershed and yield

1 Introduction

Coffee (*Coffea arabica* L.) is one of the world's most important agricultural commodities and plays a major role in Ethiopia's economy and rural livelihoods. Ethiopia is the center of origin and genetic diversity of *C. arabica*, with diverse genetic resources and agro-ecological conditions that contribute to the distinctive quality of Ethiopian coffee (Alemayehu et al., 2023). Recent studies also highlight the importance of conserving Ethiopia's coffee genetic

resources for future breeding and sustainable production (Merga et al., 2023; Aliyi et al., 2025). Coffee production in Ethiopia is dominated by smallholders, and its contribution extends beyond cash income. Coffee production systems can provide food, fuel wood, shade, soil protection, biodiversity habitat, and other ecosystem services. Recent evidence from southwestern Ethiopia shows that coffee-based agroforestry can contribute to household food security by supporting diversified production of fruits, vegetables, cereals, and other useful species (Chemeda et al., 2024). Thus, coffee technology dissemination should be understood not simply as the adoption of yield-enhancing inputs, but as part of a broader socio-ecological farming system.

Despite Ethiopia's comparative advantage in coffee production, smallholder productivity remains constrained by inadequate agronomic management, limited access to improved planting materials, weak extension and information systems, soil fertility limitations, pests and diseases, and post-harvest constraints. Adoption of agricultural technologies is influenced by socioeconomic, institutional, and informational factors rather than technology availability alone (Gemedé, K. & Adala, A. (2026). Recent evidence

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from Ethiopian coffee systems demonstrates the importance of extension education, farmer-group membership, communication technologies, and access to climate information in strengthening farmers' access to agricultural information (Kaske et al., 2023). Similarly, recent research on coffee technology adoption in the Jimma area has applied a multivariate probit framework to examine factors associated with adoption of multiple coffee technologies (Temesgen & Debeb, 2024).

Coffee agroforestry systems can maintain tree cover and contribute to biodiversity conservation, soil fertility, carbon storage, and climate resilience (Chemeda et al., 2024). Recent experimental evidence also shows that canopy and fertilizer management influence soil nitrogen dynamics in coffee agroforestry systems (Kurniawan et al., 2024). More recently, research in southern Ethiopia found that structurally complex coffee–enset agroforestry systems can maintain substantially greater carbon stocks than monocrop coffee systems (Gabiso et al., 2026).

The role of institutions and markets is equally important. Market information can affect coffee sales, marketed surplus, and coffee revenue, particularly where market participation is relatively weak (Lee et al., 2025). Recent evidence from southwestern Ethiopia further shows that coffee market supply is associated with production quantity, landholding size, livestock assets, non-farm income, and household characteristics (Yadeta et al., 2026).

The Afalach watershed provides an important case for understanding these issues because coffee production occurs within a mixed-farming and watershed-management context. However, information on the current status of coffee production technologies and farm-level management practices in the watershed is limited.

Therefore, this study aimed to assess the current status of coffee production technology dissemination in the Afalach watershed, with particular emphasis on farmers' experience, land allocation, productivity, planting materials, agronomic practices, soil fertility management, disease management, harvesting, drying, and handling practices. The study also discusses the implications of these practices for productivity, agroforestry sustainability, biodiversity, soil fertility, climate resilience, household food security, and market development.

2 Methodology

2.1 Description of the study area

Assessment of the dissemination of coffee technology was done in the Afalach watershed at Shishoende district of Kaffa Zone. The geographical location of the study area is shown

in Fig. 1. The area receives rainfall almost throughout the year, with the main rainy season extending from March to October. The mean annual rainfall is about 1877 mm, while the average minimum and maximum temperatures are 12 °C and 26 °C, respectively. The main economic activity of the local people is mixed farming system. Coffee production is a major component of the farming system and serves as the primary source of income for most households.

2.2 Research design and sampling procedure

A cross-sectional survey design was used to assess the dissemination of coffee production technologies. Both qualitative and quantitative data collection methods were applied to ensure comprehensive analysis.

A total of 113 eligible coffee-producing farmers were identified in the study watershed based on predefined eligibility criteria. After identifying eligible participants, The required sample size was determined using Yamane's formula (Yamane, 1967):

$$SS = N / [1 + N(e)^2]$$

Where: SS= Required Sample size; N= Total population; e=margin of error. (7%)

Using ($N=113$) and ($e=0.07$) (7% margin of error, equivalent to 93% precision), the calculated sample size was ($N=113/[1+113(0.07)^2]$), where based on this $SS \approx 73$, which was rounded to 73 farmers. The 73 respondents were then selected from the sampling frame using a simple random sampling technique, and primary data were collected from the selected farmers using a structured questionnaire administered through face-to-face interviews. A semi-structured questionnaire, transect walks, and field visits (home gardens, cultivated fields), key informants and focus group discussions are used for collecting information from selected farmers. Data gathered from transect walks, field visits and key informants are used for provision and confirmation of the information obtained from the semi-structured questionnaire.

2.3 Data collected

The questionnaire focused on: Household characteristics related to coffee production, Cropping systems and farm management practices, Coffee production technologies and packages used, Gender roles in coffee production, Seasonal calendar (planting, harvesting, and major constraints). Field observations were conducted to assess: Cropping patterns, Disease and insect pest incidence and Severity of production constraints.

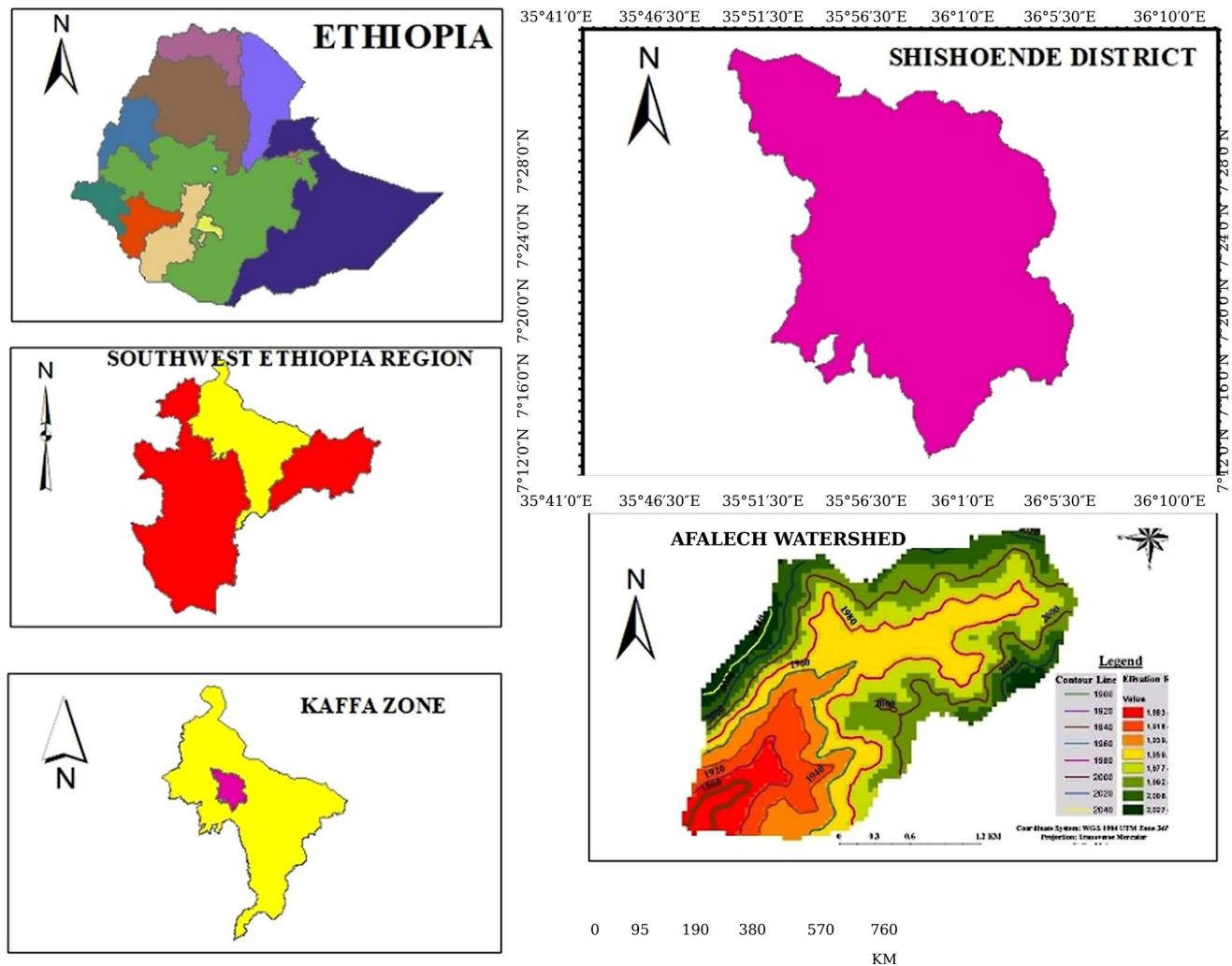


Fig. 1 Location of the Afalach watershed within Shishoende District and Kaffa Zone, southwestern Ethiopia

Table 1 Farmers experience on coffee production at Afalach watershed

Year	Frequency	Percent
1–10	50	68.49
11–20	20	27.39
21–30	1	1.36
31–40	0	0
41 and above	2	2.73
Total	73	100

2.4 Data analysis

Quantitative and qualitative social survey data collected was coded and analyzed using IBM Statistical Package for Social Science (SPSS) software, version 20 (Field, 2024). Cross-tabulation tables are constructed, and descriptive statistics was generated to summarize data from the questionnaires. To make statistical inferences, descriptive statistics, frequencies and percentages are conducted to analyze the variables.

These methods are commonly used in agricultural and social science research to interpret survey data (Field, 2024).

3 Result and discussion

3.1 Farmers experience on coffee production

The experience of farmers in coffee production is presented in Table 1. Most respondents (68.49%) had 1–10 years of coffee-farming experience, whereas only 2.73% had more than 41 years of experience. This distribution indicates that a large proportion of farmers in the Afalach watershed have relatively limited experience in coffee production, which may affect their familiarity with improved management practices and production technologies. Recent evidence from Southwest Ethiopia shows that farmer characteristics and access to production-related services are important determinants of coffee-production efficiency (Tamirat, N. &

Table 2 Land occupied with coffee plantation at Afalach watershed

land allocated	Frequency	Percent
<0.125	38	52.1
0.126 to 0.25	24	32.9
0.26 to 0.375	2	2.7
0.375 to 0.5	2	2.7
0.51 to 0.635	3	4.1
0.636 to 0.76	1	1.4
>0.76	3	4.1
Total	73	100.0

Table 3 Productivity of coffee at Afalach watershed

Productivity	Frequency	Percent
1	33	45.2
2	12	16.43
3	7	9.5
4	7	9.5
5	4	5.4
6	3	4.1
7	2	2.7
8	3	4.1
9	2	2.7
Total	73	100

Tadele, S., 2023). Similarly, Chemed et al. (2025) reported an average coffee-farming experience of about 29 years among farmers in Gimbi District, Western Ethiopia, demonstrating substantial variation in farming experience across coffee-producing areas. Therefore, the relatively shorter experience observed in the Afalach watershed may be due to variation in study areas, as some regions have a longer history of coffee cultivation than Afalach watershed.

3.2 Land allocation for coffee production

As shown in Table 2, more than half of the farmers (52.1%) allocated less than 0.125 hectares for coffee production. Only a few farmers (4.1%) used larger plots.

This indicates that coffee is produced on a small scale, despite the area being suitable for coffee production. Recent evidence indicates considerable variation in the area allocated to coffee production among Ethiopian coffee-growing households, reflecting differences in farming systems and local conditions (Tikuneh et al., 2023). Similarly, Girma Asefa and Ayalew Muluken (2024) reported that farm size is an important factor influencing the efficiency of smallholder farmers. Therefore, the predominance of small coffee plots in the study area may be associated with limited land availability and the need to allocate land to other crops. The difference may be due to better institutional support and market access in cooperative-based systems.

Table 4 Type of coffee varieties used in Afalach watershed

Type of the variety	Frequency	Percent
Local	41	56.2
Improved	32	43.8
Total	73	100

Table 5 Source of coffee varieties planted in Afalach watershed

Source of coffee seed	Frequency	Percent
Agricultural office	27	37.0
Research center	1	1.37
NGO	4	5.47
Other source	41	56.16
Total	73	100.0

3.3 Coffee productivity

Table 3 shows that 45.2% of farmers produced 1 quintal of coffee per hectare, indicating relatively low productivity in the study area compared with the reported national average of 8.4 q/ha⁻¹ (Tikuneh et al., 2023). This low productivity may be related to limited adoption of improved coffee varieties and inefficient farm management practices. Recent evidence from Ethiopia indicates that factors such as access to extension services, inputs, and improved coffee varieties influence coffee production (Tamirat & Tadele, 2023). Similarly, Tesfay (2023) reported that adoption of improved coffee varieties significantly increased coffee productivity among smallholder farmers. These findings suggest that the low productivity observed in the study area may partly reflect limited access to improved planting materials and inadequate adoption of recommended coffee production and management practices.

3.4 Coffee varieties and sources

Table 4 shows that 56.2% of farmers used local coffee varieties, compared with 43.8% who used improved varieties, indicating relatively low adoption of improved varieties in the study area. Recent studies in Ethiopia found that adoption of improved coffee varieties can increase productivity and is influenced by access to extension services, training, and improved planting materials (Tesfay, 2023). The relatively high use of local varieties in the study area may therefore be associated with limited access to improved planting materials and support services.

The source of coffee varieties planted in the Afalach watershed is presented in Table 5. More than half of the respondents (56.16%) could not identify the source of their coffee varieties, while 37.0% obtained them from agricultural offices, 5.47% from NGOs, and 1.37% from research centers. This pattern suggests limited farmer awareness and gaps in the dissemination of improved planting materials. Recent studies have similarly identified limited access to

improved coffee varieties and extension support as important constraints to their adoption in Ethiopia (Tesfay, 2023; Afeto & Belay, 2025).

3.5 Use of coffee production packages

Table 6 indicates that 68.49% of farmers did not use the recommended coffee production packages, indicating low adoption of improved coffee technologies in the study area. Recent evidence from southwestern Ethiopia shows that the adoption of improved coffee varieties and recommended management practices remains limited and is influenced by institutional and resource-related factors (Temesgen & Debeb, 2024).

3.6 Farm management practices

3.6.1 Coffee slashing, spacing, and hole preparation

Tables 7, 8, 9 and 10 show considerable variation in slashing time, planting spacing, and hole preparation, with many farmers not following the recommended practices. This indicates low adoption of recommended coffee management technologies in the study area. Recent studies shows that adoption of improved coffee technologies and agro-nomic practices remains limited and is influenced by farmers' access to extension services, technical knowledge, and production resources (Temesgen & Debeb, 2024; Gada et al., 2026).

Intra and inter row spacing during plantation in the watershed was studied and presented in Table 8 below. As the interview result revealed that, the coffee planting spacing in the watershed ranged from 1*1m to 2*2m. 7 (9.6%) of the respondents used 1.5*1.5 m, 6 (8.2%) planted their coffee by 1.8*1.8 m, 13 (17.8%) of the respondent used 1*1 meter, 15 (20.6) of the respondents planted their coffee with the spacing of 2*2 and the rest 32 (43.8%) planted without measuring (Table 8).

Calendar for hole digging in the watershed was interviewed and presented in Table 9. As the result implied that time for hole digging varies from farmer to farmer. 16 (21.9%) of the respondents conducted hole digging from February to March, 19 (26%) farmers dig the hole from April to May, 9 (12.3%) of the respondents dig the hole on January, 1 (1.4%) respondent dig the on July and the rest 28 (38.35%) doesn't had schedule (Table 9).

The depth and width of the hole for coffee plantation in Afalach watershed was shown in Table 10. The result implies that, from the total respondents 73 (100%), 45 (61.6%) used 30*30 cm, 18 (24.7%) of the farmers prepared the hole with depth and width of 40*40 cm, 6 (8.2%) of the respondents used 50*50 cm, and 3 (4.1%) of the respondent

Table 6 Assessment of implementation of coffee production packages

Coffee production packages	Frequency	Percent
Yes	23	31.5
No	50	68.49
Total	73	100

Table 7 Slashing calendar for new plantation coffee in Afalach watershed

Slashing calendar	Frequency	Percent
Jan and Feb	17	23.3
Feb and March	10	13.7
March and April	9	12.3
April and May	3	4.1
Un known	34	46.6
Total	73	100

Table 8 Intra and inter spacing of coffee in Afalach watershed

Intra and inter spacing	Frequency	Percent
1*1	13	17.8
1.5*1.5	7	9.6
1.8*1.8	6	8.2
2*2	15	20.6
Not measured	32	43.8
Total	73	100

Table 9 Calendar for hole digging in Afalach watershed

Calendar for hole digging	Frequency	Percent
February-March	16	21.9
April-May	19	26.0
January	9	12.3
July	1	1.4
Not scheduled	28	38.35
Total	73	100

Table 10 The depth and width of the hole for coffee plantation in Afalach watershed

Depth and width	Frequency	Percent
30*30	45	61.6
40*40	18	24.7
50*50	6	8.2
60*60	3	4.1
Without measuring	1	1.4
Total	73	100

used 60*60 cm, while 1.4% did not measure the hole dimensions (Table 10).

3.6.2 Planting time

Table 11 shows considerable variation in planting time, with farmers planting from March to July and the highest proportions in July (30.13%) and April (27.40%), indicating absence of a clear planting calendar. This is important because recent studies in southwestern Ethiopia show that

Table 11 Seedling plantation time in Afalach watershed

Months	Frequency	Percent
March	16	21.92
April	20	27.4
May	9	12.33
Jun	6	8.22
July	22	30.13
Total	73	100

Table 12 Weed controlling methods from coffee farm in Afalach watershed

Weed controlling methods	Frequency	Percent
Slashing	60	82.2
Mulching	9	12.3
By using cover crops	4	5.5
Total	73	100

Table 13 Pruning activity in Afalach watershed on coffee farm

Pruning practice	Frequency	Percent
No	55	75.3
Yes	18	24.7
Total	73	100

Table 14 Diseases assessment practice of coffee producers in Afalach watershed

Diseases assessment practice	Frequency	Percent
No	28	38.4
Yes	45	61.6
Total	73	100

coffee seedlings are sensitive to soil-moisture stress during establishment (Asefa, 2023; Asefa et al., 2025). Therefore, planting seedlings when rainfall is reliable and soil moisture is adequate should be emphasized through agricultural extension services.

3.6.3 Weed control, pruning, and disease management

Weed control was dominated by slashing (82.2%), while mulching (12.3%) and cover crops (5.5%) were rarely used (Table 12); similarly, only 24.7% of farmers practiced pruning (Table 13). This limited adoption of improved management practices may constrain coffee productivity. This result is consistent with the finding of Daba et al. (2023), who reported the negative effects of weeds on coffee yield, and Mohammed and Worku (2025), who found that appropriate pruning significantly improved coffee yield and yield components in southwest Ethiopia.

The knowledge of stakeholders regarding coffee diseases was assessed and presented in Table 14 below. Diseases is one of yield reduction factor. From total of 73 (100%) respondents 45 (61.6%) of the respondent assess diseases and reported for respective bodies. The rest 28 (38.4%)

Table 15 Fertilizer utilization status of farmer in Afalach watershed

Organic Fertilizer	Frequency	Percent	Inorganic Fertilizer	Frequency	Percent
No	52	71.2	no	73	100
Yes	21	28.8	yes	0	0
Total	73	100	Total	73	100

Table 16 Coffee harvesting method in Afalach watershed

Harvesting method	Frequency	Percent
Selective picking	71	97.3
Collecting of failed cherries	2	2.7
Total	73	100

farmers never assess diseases from their coffee plantation farm (Table 14).

3.6.4 Fertilizer use

Fertilizer use was generally low in the Afalach watershed: 71.2% of farmers did not apply organic fertilizer, while only 28.8% used it, and all farmers reported that they did not use inorganic fertilizer (Table 15). This limited use of soil-fertility inputs may constrain coffee productivity, as Gebisa (2024) reported that combining coffee-husk organic fertilizer with mineral fertilizer significantly improved coffee yield in southern Ethiopia. Similarly, Takala and Chali (2025), emphasized the importance of integrated soil-fertility management for improving coffee productivity in Ethiopia.

3.7 Harvesting and post-harvest practices

3.7.1 Harvesting practices

As shown in Tables 9 and 16, 167.3% of farmers practiced selective picking, while only 2.7% collected fallen or unripe cherries. This high use of selective harvesting is important because picking ripe cherries helps reduce defects and maintain coffee quality.

Banti, & Atlaw, 2024, also emphasized the importance of using fully matured red-ripe cherries in coffee quality evaluation in Jimma, Ethiopia. Thus, the high level of selective picking in the Afalach watershed indicates good adoption of a quality-oriented harvesting practice.

3.7.2 Drying practices

As shown in Table 17, most farmers (68.5%) dried coffee on locally constructed raised beds, while 21.9% dried it directly on the ground, 2.7% used wire-mesh beds, and 6.8% sold fresh cherries without drying. Raised-bed drying is preferred because it improves air movement, reduces contact with soil and other contaminants, and helps achieve

Table 17 Coffee drying materials in Afalach watershed

Drying materials types	Frequency	Percent
On ground	16	21.9
On locally constructed bed	50	68.5
no drying	2	2.7
Total	5	6.8
	73	100

Table 18 Coffee handling materials in Afalach watershed

Coffee bean handling material	Frequency	Percent
Sack	32	43.8
Bag	22	30.1
plastic	10	13.7
Other	9	12.3
Total	73	100

Table 19 Shows coffee production system in Afalach watershed

Coffee production systems	Frequency	Percent
Garden coffee(inter cropping with enset and fruit crops)	50	68.5
Plantation coffee plots	3	4.1
Forest coffee	5	6.8
Semi forest coffee	15	20.5
Total	73	100

more even drying. Recent studies in Ethiopia also show that drying conditions and methods can affect coffee quality and safety (Abdissa et al., 2023; Banti & Atlaw, 2024). Therefore, the relatively high use of raised beds in the Afalach watershed is encouraging, although the continued use of ground drying may increase the risk of contamination and uneven drying.

3.7.3 Coffee handling practices

The study also examined materials used for handling and storage (Table 18). Sacks were the main coffee-handling material (43.8%), followed by bags (30.1%), plastic (13.7%), and other materials (12.3%) (Table 18). This result agrees with Ummeta (2025), who reported widespread use of jute sacks in southwestern Oromia. However, Eshete et al. (2024) found that hermetic PICS and GrainPro bags preserved coffee quality better than jute sacks during storage.

In contrast, Gutema et al. (2025) reported that 70.8% of farmers in Ari and South Omo used polyethylene bags, with 49.1% experiencing mold during storage. These differences may be related to packaging availability and storage conditions. Promoting clean, moisture-protective packaging, especially hermetic bags, could therefore help maintain coffee quality in the Afalach watershed.

Table 20 Coffee marketing channels used by farmers in the Afalach watershed

Coffee market chanel	Frequency	Percent
Local collectors/traders	44	60.3
Cooperatives	10	13.7
Wholesalers	0	0
Local consumers	5	6.8
Private washing stations	14	19.2
Total	73	100

3.8 Coffee production systems and agroforestry context

Table 19 shows that garden coffee was the dominant production system (68.5%), followed by semi-forest coffee (20.5%), forest coffee (6.8%), and plantation coffee (4.1%). The dominance of garden coffee indicates that farmers commonly grow coffee with enset and fruit crops around their homes. This agrees with Urugo et al. (2025), who reported that garden coffee is widely practiced in Ethiopia as a diversified agroforestry system. Similarly, Kebebew and Ozanne (2024) highlighted the importance of forest-based coffee systems in southwestern Ethiopia. The low proportion of plantation coffee suggests that intensive, single-crop coffee production is less common in the watershed. Overall, coffee production in Afalach is mainly based on diversified agroforestry systems, which can support both farm livelihoods and environmental conservation.

3.9 Coffee marketing and market information

Table 20 shows that local collectors/traders were the main coffee marketing channel (60.3%), followed by private washing stations (19.2%), cooperatives (13.7%), and local consumers (6.8%), while none sold directly to wholesalers. The high dependence on local traders indicates limited use of alternative market outlets. This finding agrees with Mihret (2026), who examined coffee farmers' market outlet choice in Gimbo Woreda, Kaffia Zone, Southwest Ethiopia, and showed that farmers' choice of market outlet is influenced by factors related to market access and farmer characteristics. Therefore, strengthening cooperatives and improving farmers' access to market information and direct buyers could help increase their bargaining power and marketing benefits.

4 Conclusion and recommendations

The study area has favorable environmental conditions for coffee production. However, the overall productivity is very low due to several factors. These include limited farming experience, small land allocation, low use of improved

varieties, poor adoption of recommended practices, lack of proper training, and weak extension support.

Most farmers rely on traditional methods and do not follow recommended agronomic practices such as proper spacing, pruning, fertilizer use, and disease management. In addition, the absence of a clear production calendar and limited awareness of improved technologies further reduce productivity.

To improve coffee production in the area, a combined approach is needed. Farmers should be supported through regular training on improved coffee production practices, including pruning, compost preparation, and disease control. Improved coffee varieties should be widely distributed and demonstrated under local conditions to increase adoption.

Extension services need to be strengthened to provide clear guidance on planting schedules, spacing, and farm management practices. Awareness programs should also be conducted to improve farmers' knowledge of pests and diseases.

Furthermore, promoting the use of organic fertilizers and improving post-harvest handling practices will help increase both yield and quality. Coordinated efforts among agricultural offices, research institutions, and development organizations are essential to bring sustainable improvements in coffee production.

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Data availability The collected data and analysis in this study can be obtained from the corresponding author upon reasonable request.

Declaration

Ethics approval and consent to participate This study was conducted in accordance with Ethiopia's National Research Ethics Review Guideline (Fifth Edition), the Ethiopian Institute of Agricultural Research (EIAR) Code of Research Ethics (2014), and the established research proposal review and approval protocol of the Southwest Ethiopia Agricultural Research Institute (SWEARI). The study was a non-clinical, minimal-risk agricultural field survey involving adult coffee-producing farmers and concerned agricultural production practices, technology dissemination, and related socioeconomic information. It did not involve clinical procedures, medical interventions, biological samples, human tissue, or sensitive personal or health information. The research proposal was reviewed and approved for implementation by the SWEARI Proposal Review Forum during 15–18 May 2023. For this specific non-clinical agricultural field survey, the SWEARI Proposal Review Forum formally determined that separate review by a Research Ethics Committee (REC), Institutional Review Board (IRB), or Ethical Review Board (ERB) was not required under the applicable institutional research review protocol. The Proposal Review Forum

was the institutional body responsible for review and authorization of the research proposal; it was not constituted as a separate REC/IRB/ERB, and therefore no REC/IRB/ERB approval number was issued. Before data collection, the study objectives and procedures were explained to all participants, and voluntary informed consent was obtained from each respondent. All participants were adults aged 24–58 years, and none belonged to a vulnerable group. Participation was voluntary, respondents were free to decline participation, and confidentiality and anonymity were maintained. The information collected was used solely for research purposes.

Competing interests The authors declare no competing interests.

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